Resumen-Montero-2025

Title

A simple physical model for glacial cycles.

Volume treatment in PACCO

Pérez_Montero_etal_2025_pacco, p.18

> Since PACCO has no spatial dimensions, the ice volume is defined through the product of ice thickness and the potentially glaciated surface S. This can be expressed in meters of sea level equivalent (m SLE) by
>
> $$V_{ice} = \frac{\rho_{\text{ice}}}{\rho_{\text{wtr}} \cdot S_{\text{ocn}}} \cdot S(T, v_b) \cdot H,$$
>
> $$S = 8 \cdot \pi \cdot L^2 \cdot \left(\frac{T - T_{\text{ref}}}{A_{\text{thr}}} - \frac{v_b}{v} \right).$$

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> This equation relies on the fact that a theoretical ice sheet is a symmetrical dome (π · L2) whose extension can be modified by both anomalies in regional climate (T − Tref) · A−1 thr and excessive sliding velocities vb · u−1. Note that Athr is the thermal amplitude provided by a certain extent of the ice sheet, u is the typical velocity in an ice stream and **S is multiplied by 8 that is a free constant parameter to obtain similar amplitude to proxies.**

This is a bit gross of approximation, isn't it?

Introduction

Pérez_Montero_etal_2025_pacco, p.4

> We have built a model with the minimum amount of explicitly-resolved physical processes that successfully reproduces the GIV of the Pleistocene. Our model is an efficient and spatially adimensional climate-cryosphere model with explicitly-solved physical processes that can be enabled or disabled independently, allowing us to isolate the physical processes that control the mechanisms underlying the Pleistocene records.

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> Here we describe the **Physical Adimensional Climate-Cryosphere mOdel (PACCO)** in a progressive manner in order to provide a full picture of its capacities together with the physical basis for its formulation. PACCO represents conceptually the interaction between climate and Northern Hemisphere ice sheets using a system of up to 6 coupled ODEs (Ordinary Differential Equations) as described in the following sections.

Model description and results

Table 1. Summary of the experiments performed in this work. Experiments are ordered in gradual increasing complexity.

Experiment	Description	Section
LIN	SIF, ISD, linear SMB	2.1
NONLIN	SIF, ISD, non-linear SMB	2.2
ISOS	SIF, ISD, isostatic response	2.3
RISOS	RIF, ISD, isostatic response	2.4
BASE	RIF, ISD, carbon cycle, albedo	2.5
THERM	RIF, ISD, carbon cycle, albedo, IST	2.6
AGING	RIF, ISD, carbon cycle, albedo aging	2.7

SIF = Synthetic Insolation Forcing, RIF = Real Insolation Forcing, ISD = Ice Sheet Dynamics, IST = Ice Sheet Thermodynamics, SMB = Surface Mass Balance

Table 2. Parameters from LIN, NONLIN, ISOS and RISOS experiments. Note that the parameters not referenced correspond to model calibration values.

Parameter	Value	Parameter	Value	Reference
t_0 (kyr BP)	800	I_{ref} ($\mathrm{W \cdot m^{-2}}$)	480	Present-day anomaly
t_f (kyr BP)	0 (J2000)	P_p, P_o, P_e	0.6, 0.4, 0.0	Paleoclimatic constraint
T_0 (^{o}C)	0	τ_p, τ_o, τ_e (kyr)	23, 41, 100	Paleoclimatic constraint
H_0 (m)	0	T_{ref} (^{o}C)	0	Present-day anomaly
B_0 (m)	500	A_T (K)	15	Paleoclimatic constraint
ρ_{bed} ($\mathrm{kg \cdot m^{-3}}$)	2700	I_{min} ($\mathrm{W \cdot m^{-2}}$)	425	Paleoclimatic constraint
ρ_{ice} ($\mathrm{kg \cdot m^{-3}}$)	910	I_{max} ($\mathrm{W \cdot m^{-2}}$)	565	Paleoclimatic constraint
λ ($\mathrm{m{\cdot}yr^{-1}{\cdot}K^{-1}}$)	0.2	L (km)	1000	Oerlemans et al. (2008)
T_{thr} (^{o}C)	-5	f_{str}	0.2	Margold et al. (2015)
$\dot{s}_{\mathrm{ref}}$ ($\mathrm{m \cdot yr^{-1}}$)	0.2	A_f ($\mathrm{Pa^{-3}{\cdot}yr^{-1}}$)	10^{-16}	Glen (1958)
$k_{\dot{s}}$ ($\mathrm{m \cdot yr^{-1} \cdot K^{-1}}$)	0.0015	C_s ($\mathrm{m{\cdot}yr^{-1}{\cdot}Pa^{-2}}$)	10^{-10} to 10^{-4}	Pollard and DeConto (2012)
B_{ref} (m)	500	τ_B (yr)	5000	Le Meur and Huybrechts (1996); Swierczek-Jereczek et al. (2023)

Table 3. Parameters from BASE, THERM and AGING experiments. Note that the parameters not referenced correspond to model calibration values.

Parameter	Value	Parameter	Value	Reference
t_0 (kyr BP)	900	T_{ref} (^{o}C)	0	Present-day anomaly
t_f (kyr BP)	0 (J2000)	S_{ocn} (km^2)	$3.618{\cdot}10^8$	Cogley (2012)
T_0 (^{o}C)	0	α_{l}, α_{ni}, α_{oi}	0.2, 0.9, 0.25	Cuffey and Paterson (2010)
H_0 (m)	0	I_{ref} ($\mathrm{W \cdot m^{-2}}$)	480	Present-day anomaly
B_0 (m)	500	L (km)	1000	Oerlemans et al. (2008)
C_0 (ppm)	280	C_{ref} (ppm)	280	Present-day anomaly
A_0 (yr)	0	C_s ($\mathrm{m{\cdot}yr^{-1}{\cdot}Pa^{-2}}$)	10^{-10} to 10^{-4}	Pollard and DeConto (2012)
$f_{str,0}$	0.0	$A_{\mathrm{thr}}(K)$	20	Paleoclimatic constraint
$T_{ice,0}$ (^{o}C)	-15	f_{str}	0.2	Margold et al. (2015)
ρ_{bed} ($\mathrm{kg \cdot m^{-3}}$)	2700	k_I ($\mathrm{m \cdot yr^{-1} \cdot W^{-1} \cdot m^{-2}}$)	0.027	
ρ_{ice} ($\mathrm{kg \cdot m^{-3}}$)	910	λ ($\mathrm{m{\cdot}yr^{-1}{\cdot}K^{-1}}$)	0.1	
ρ_{wtr} ($\mathrm{kg \cdot m^{-3}}$)	1000	T_{thr} (^{o}C)	-5	
τ_T (yr)	900	$\dot{s}_{\mathrm{ref}}$ ($\mathrm{m \cdot yr^{-1}}$)	0.3	
k_T ($\mathrm{J \cdot s^{-1} \cdot m^{-1} \cdot K^{-1}}$)	2.1	$k_{\dot{s}}$ ($\mathrm{m \cdot yr^{-1} \cdot K^{-1}}$)	0.0015	
c_I ($\mathrm{K \cdot m^2 \cdot W^{-1}}$)	0.1	ν ($\mathrm{m{\cdot}yr^{-1}}$)	300	
c_C ($\mathrm{K \cdot m^2 \cdot W^{-1}}$)	0.65	A_f ($\mathrm{Pa^{-3}{\cdot}yr^{-1}}$)	10^{-16}	Glen (1958)
c_z ($\mathrm{K \cdot m^{-1}}$)	0.007	B_{ref} (m)	500	
$k_{T,C}$ ($\mathrm{ppm \cdot K^{-1}}$)	5	H_b (m)	10	Robel et al. (2013)
h_{geo} ($\mathrm{W{\cdot}m^{-2}}$)	10^{-3} to 10^{-2}	τ_{kin} (yr)	10 to 10^4	Payne et al. (2004); Nick et al. (2009)
τ_C (yr)	10	$f_{\mathrm{str,min}}$	0.0	
$f_{\mathrm{str,max}}$	0.5	τ_α (yr)	10^3 to 10^6	
k_α ($\mathrm{yr^{-1}}$)	$5{\cdot}10^{-6}$	τ_B (yr)	5000	Le Meur and Huybrechts (1996); Swierczek-Jereczek et al. (2023)

(Ice sheet thermodynamics and dynamics equations are the same for all experiments, and are described before the experiment sections).

LIN Experiment

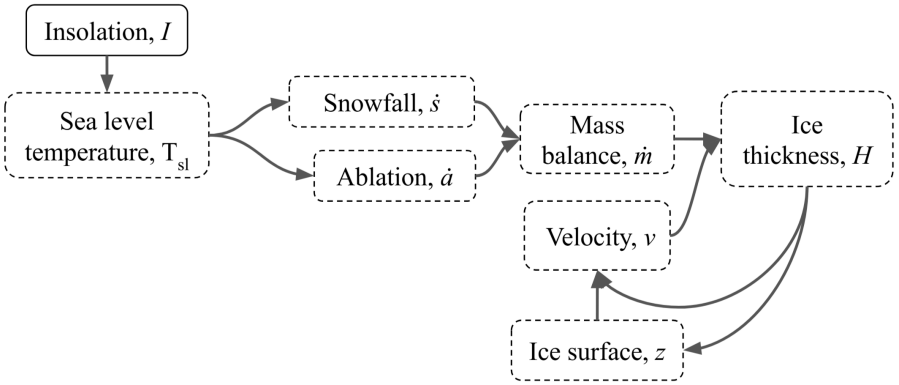


Figure 2. LIN experiment scheme. Note the linear relationship between the forcing and the evolution of the ice thickness.

Pérez_Montero_etal_2025_pacco, p.6

> We start by building a simple configuration that represents the interaction between insolation and ice-sheet dynamics. The simplest configuration of PACCO receives insolation as the only forcing and, for that purpose, we defined a synthetic insolation forcing as a linear combination of three cosines with periods identical to those of orbital parameters

Pérez_Montero_etal_2025_pacco, p.6

> Insolation is then translated to sea-level temperature via
>
> $$T_{\mathrm{sl}} = T_{\mathrm{ref}} + A_T \cdot \bar{I},$$
>
> $$\bar{I} = 2 \cdot \frac{I - I_{\min}}{I_{\max} - I_{\min}} - 1,$$
>
> that is, the normalized insolation between -1 and 1.

Pérez_Montero_etal_2025_pacco, p.7

> The “climate response” is fed into the ice sheet through the surface mass balance ˙m. This is assumed to be the only source of mass balance of the ice sheet (i.e,

basal mass balance and calving are ignored) with the exception of the ice discharge (Eq. 3). Thus,

$$\dot{m} = \dot{s} - \dot{a},$$

where the dot indicates rate of ice mass change ($\mathrm{m} \cdot \mathrm{yr}^{-1}$); hence $\dot{s}$ is snowfall and $\dot{a}$ is ablation.

And snowfall rate is represented as a linealization of Clausius-Clapeyron:

$$\dot{s} = \dot{s}_{\mathrm{ref}} + k_{\dot{s}} \cdot \left(T_{\mathrm{sl}} - T_{\mathrm{ref}} \right),$$

The ablation term ˙a in Eq. (15) follows a similar approach to the Positive Degree Day method (Braithwaite, 1980; Reeh, 1991; Ritz et al., 1996; Cuffey and Marshall, 2000; Huybrechts et al., 2004;165 Charbit et al., 2008; Robinson et al., 2010) that depends on the number of days in a year with sea-level temperature above the melting point. PACCO assumes a linear relationship between the ablation and the temperature anomaly and reduces the melting point constraint to Tthr. In this way, we can allow for melting at lower temperatures to account for the lack of spatial and temporal knowledge. We then define ablation as:

$$\dot{a} = \lambda \cdot \left(T_{\mathrm{sl}} - T_{\mathrm{thr}} \right).$$

Here, λ is a parameter that transfers temperature anomaly to ice mass loss.

Pérez_Montero_etal_2025_pacco, p.8

> This configuration essentially consists of a linear relation between insolation and mass balance and a nonlinear one between ice thickness and velocity.

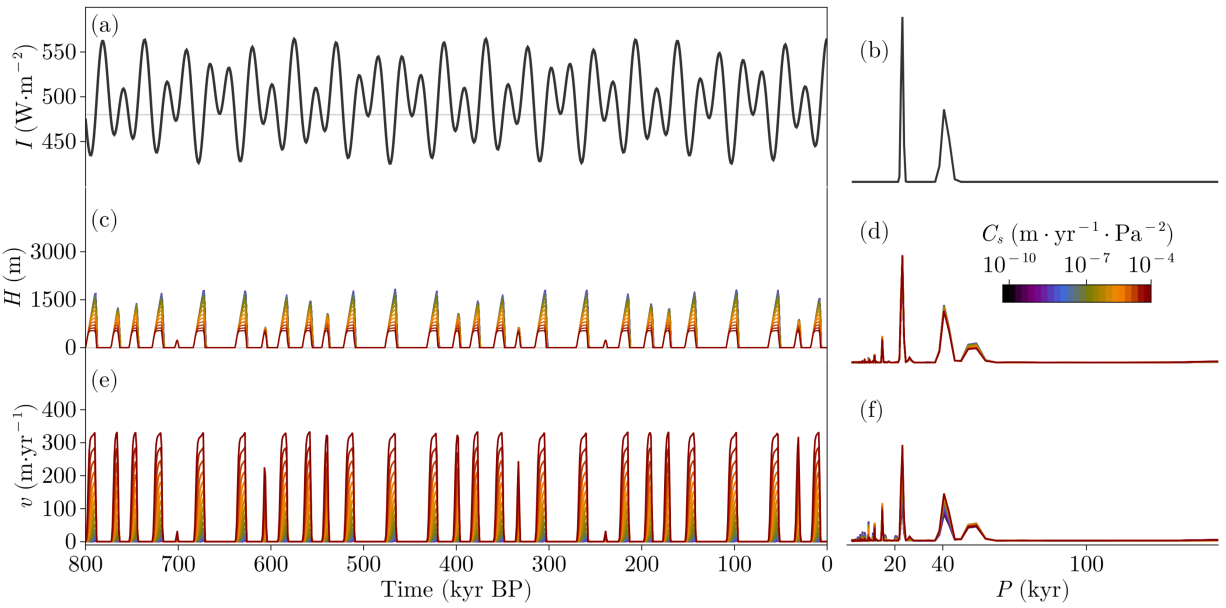


Figure 3. Results of the LIN simulation. (a, c, e) Time series obtained from the model using different sliding factors C_s. (b, d, f) Periodograms obtained from the time series in the left column. Note that when normalizing spectra, series were cut off for periods larger than 200 kyr.

Pérez_Montero_etal_2025_pacco, p.8

> As expected, when integrating over 800 kyr, glacial inceptions and deglaciations respectively occur shortly after insolation minima and maxima with a pattern that resembles the sawtooth observed in proxy records (Fig. 3). Interestingly, the power spectrum of the response of H and v presents a peak around periodicities of 60 kyr, which is however absent from the forcing. This is the manifestation of the nonlinearities introduced by the ice dynamics and is the consequence of the fact that certain interglacials are very long, since the system evolves explicitly with insolation and only those insolation minima below a certain threshold allow for a positive mass balance, leading to an increase of the ice mass. In short, the ice-sheet dynamics creates a certain degree of nonlinearity that appears, however, to be not powerful enough to allow for a realistic GIV simulation.

NONLIN Experiment

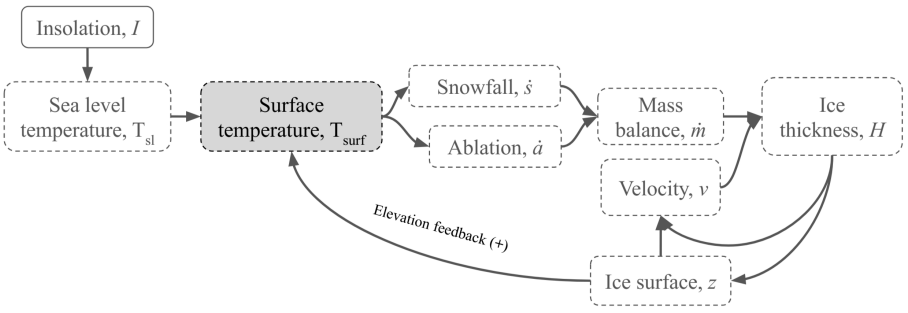


Figure 4. NONLIN experiment scheme. The previously linear relationship between the forcing and the evolution of the ice thickness (LIN) is now nonlinear due to the elevation feedback. Model additions with respect to LIN are highlighted.

Pérez_Montero_etal_2025_pacco, p.8

> The ice-surface elevation feedback (Weertman, 1961; Clark and Pollard, 1998; Oerlemans, 2003) is known to be a fundamental process controlling ice-sheet accumulation and ablation. The elevation feedback accelerates melting under a shrinking ice sheet and limits ablation when the ice sheet is growing, leading to hysteresis in the ice sheet's volume with respect to temperature (Robinson et al., 2012; Garbe185 et al., 2020). To take this feedback into account, the mass balance equations are modified by using Tsurf instead of Tsl, where

$$T_{\mathrm{surf}} = T_{\mathrm{sl}} - \Gamma \cdot z$$

$$\dot{s} = \dot{s}_{\mathrm{ref}} + k_{\dot{s}} \cdot (T_{\mathrm{surf}} - T_{\mathrm{ref}}),$$

$$\dot{a} = \lambda \cdot (T_{\mathrm{surf}} - T_{\mathrm{thr}}),$$

where $\Gamma = 0.0065 \quad \mathrm{K} \cdot \mathrm{m}^{-1}$ is the atmospheric lapse rate.

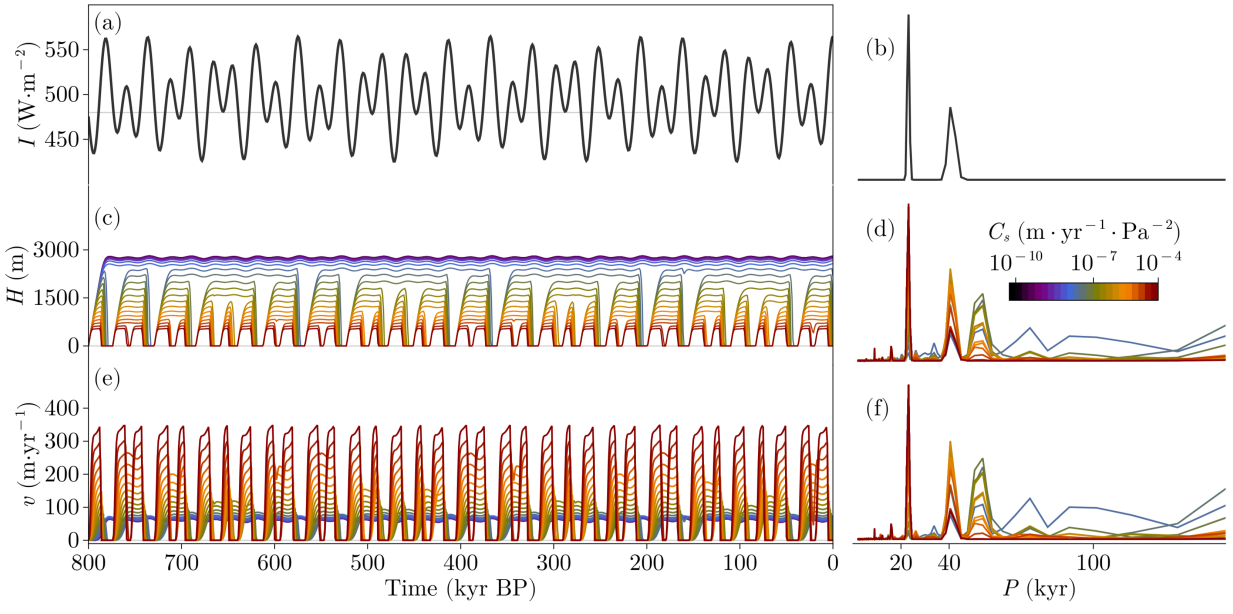


Figure 5. Results of the NONLIN simulation. (a, c, e) Time series obtained from the model using different sliding factors C_s and (b, d, f) associated periodograms. Note that when normalizing spectra, series were cut off for periods larger than 200 kyr.

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> Introducing the elevation feedback in the mass balance (Fig. 4) clearly alters the response to the forcing (Fig. 5). If the ice sheet is dynamic enough (i.e. high values of the sliding factor Cs), the model resonates to certain multiples of the insolation fundamental periods. This means that the elevation feedback introduces a non linearity via the modulation of the height amplitude, which increases the inertia of the system (Abe-Ouchi et al., 2013). Therefore, as in the previous formulation (LIN), periodicities of 60 kyr can be found, but also 80, 100 and 120195 kyrs peaks now emerge. Still, this phenomenon does not produce a dominant GIV periodicity of 100 kyr.

ISOS Experiment

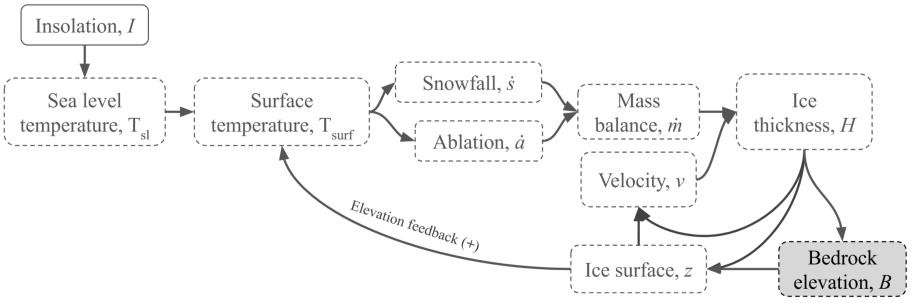


Figure 6. ISOS experiment scheme. Note that the surface elevation feedback is altered due to bedrock response. Model additions with respect to NONLIN are highlighted.

Pérez_Montero_etal_2025_pacco, p.9

> Changes in ice load lead to delayed vertical bedrock motion, a process commonly known as isostatic adjustment. This effect is included here via another prognostic variable of the model (Fig. 6), such that the bedrock elevation B evolves according to
>
> $$\frac{dB}{dt} = \frac{\left[B_{\mathrm{ref}} - \frac{\rho_{\mathrm{ice}}}{\rho_{\mathrm{bed}}} \cdot H\right] - B}{\tau_B},$$
>
> where Bref is a model parameter representing the elevation of Earth's surface in the absence of ice, ρbed is the bedrock density (c.f. Table 2), and τB is the relaxation time of the bed.

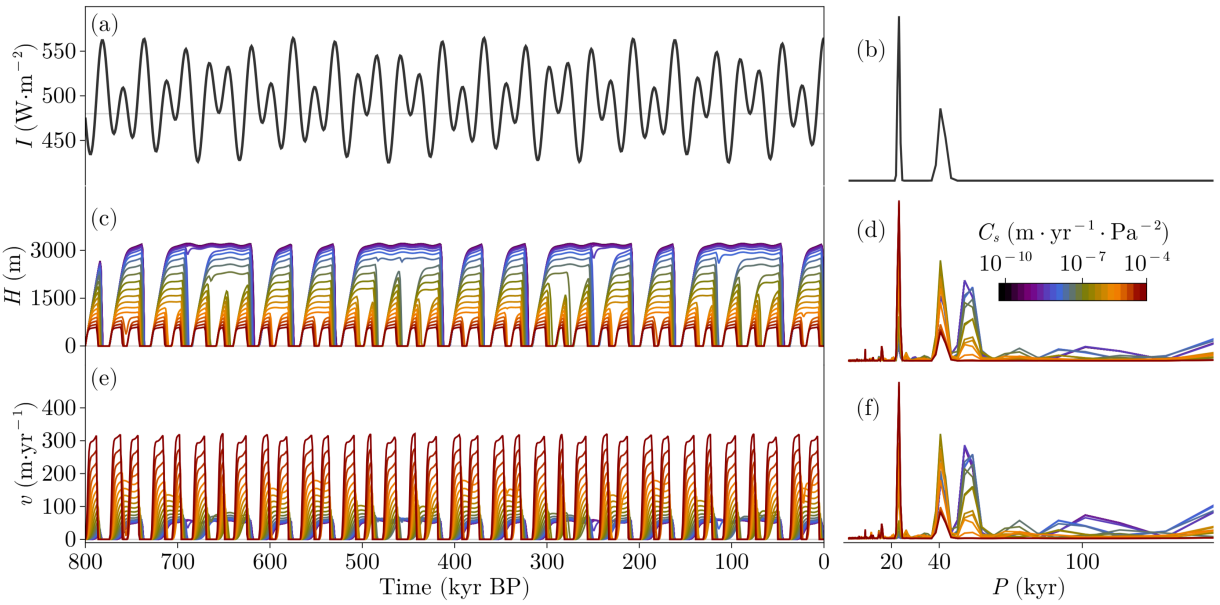


Figure 7. Results from the ISOS simulation. (a, c, e) Time series obtained from the model using different sliding factors C_s and (b, d, f) associated periodograms. Note that when normalizing spectra, series were cut off for periods larger than 200 kyr.

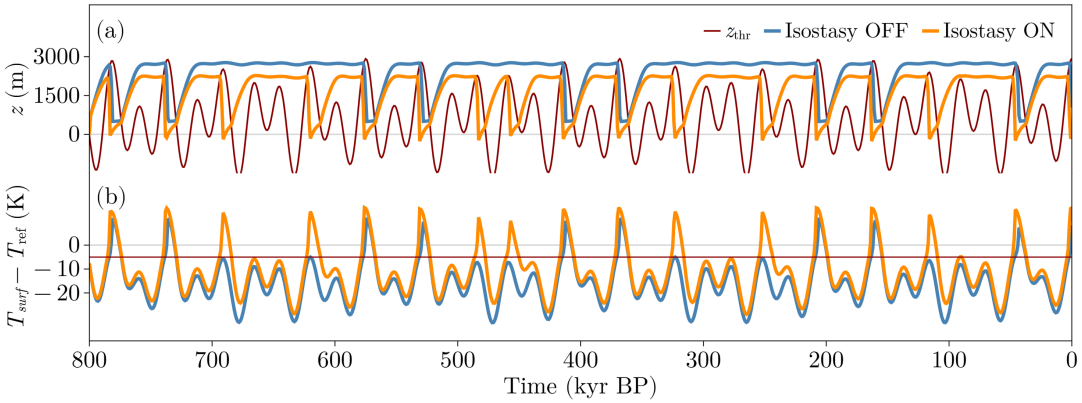


Figure 8. Time series for (a) ice surface elevation and z_{thr} and (b) surface temperature anomaly (with respect T_{ref}). z_{thr} is the point where $\dot{a}$ surpasses $\dot{s}$, and the horizontal red line indicates T_{thr} which is the isoline where $\dot{a}$ becomes positive. Note that the plotted curves are for the same sliding factor $C_s = 5.8 \cdot 10^{-8}\,\mathrm{m \cdot yr^{-1} Pa^{-2}}$.

Pérez_Montero_etal_2025_pacco, p.10

> Resonance of the system to longer periods is favored for the less dynamic runs (lowest Cs values). Reaching a glacial termination is now facilitated by the fact that ablation is greater than accumulation at low elevations (Fig. 8). We can define an elevation threshold zthr such that when z ≤ zthr (i.e. z is located below the red curve in Fig. 8) ablation is surpassed by accumulation. In other words, zthr represents the equilibrium line altitude in the ice sheet
>
> $$z_{\mathrm{thr}} = \frac{\dot{s}_{ref} + k_{\dot{s}} \cdot (T_{sl} - T_{ref}) - \lambda \cdot (T_{sl} - T_{thr})}{(k_{\dot{s}} - \lambda) \cdot \Gamma}.$$

[!PDF|red] Pérez_Montero_etal_2025_pacco, p.11

> This happens more easily for the ISOS experiments since the simulated ice-sheet heights are generally lower for the same set of parameters, as a consequence of the bedrock depression induced by the ice load. Thus, isostasy favors glacial terminations and 100 kyrs cycles can now more clearly emerge for certain values of the sliding factor. We must bear in mind, however, that210 so far our insolation forcing is synthetic. We therefore next turn to a more realistic forcing to assess whether this formulation allows for a good match with paleodata

RISOS Experiment

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> To answer if the only missing piece is the form of the forcing we perform the RISOS experiment with the ISOS configuration (Fig. 6) but using the boreal summer solstice insolation at 65ºN (Fig. 9) as forcing I (obtained following Berger, 1978; Laskar215 et al., 2004).

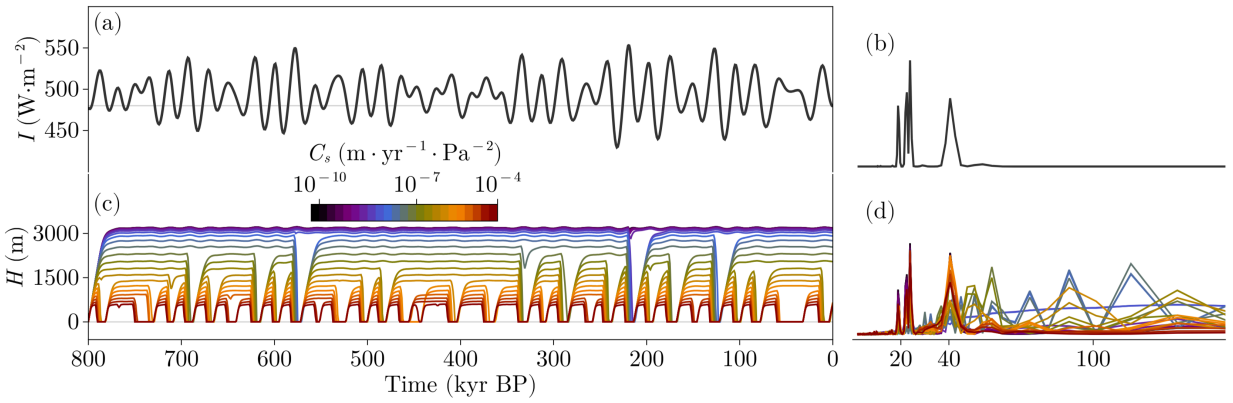


Figure 9. Results of the RISOS simulation. (a, c, e) Time series obtained from the model using different sliding factors C_s. (b, d, f) Periodograms obtained from the time series in the left column. Note that when normalizing spectra, series were cut off for periods larger than 200 kyr.

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> The model produces resonance at higher periodicities than the ones corresponding to obliquity and precession, so glacial terminations occur at different multiples of the former, depending on the sliding intensity (Fig. 9). However, the simulated timing does not match yet that provided by paleoclimatic proxies. The model response is still quite linear with the insolation forcing, so the amplitude of some of its peaks is not enough to make the ice sheet deglaciate (for low sliding) or, on the contrary, a moderate increase in insolation easily induces a termination (for high sliding). The absence of a satisfactory synchronization with the observed deglaciations, even for

mid-range values of the sliding factor, likely indicates that the model still lacks some important climatic processes. These will be addressed in the next section.

BASE Experiment

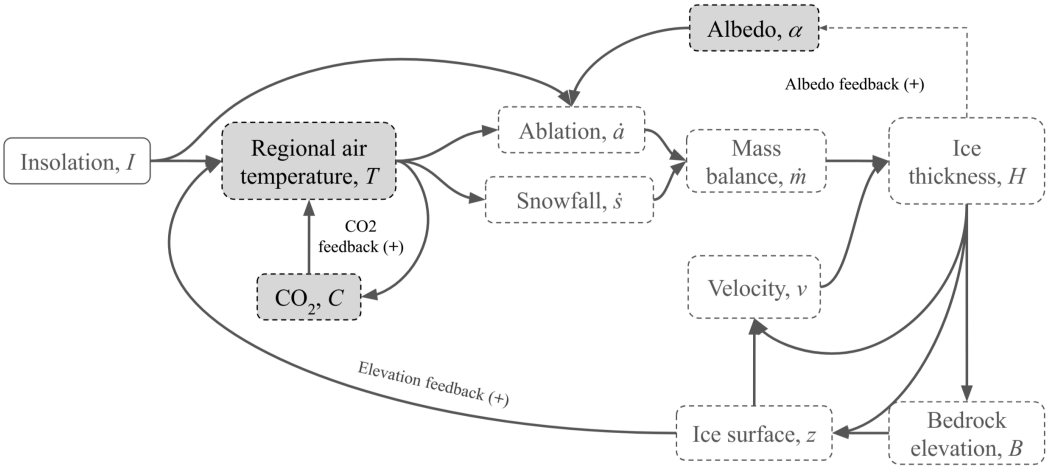


Figure 10. BASE experiment scheme. Model additions with respect to ISOS are highlighted.

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The first required improvement concerns the treatment of air temperature in the model, which is now regionalized in order to include the two-way interaction between the atmosphere and the cryosphere, as well as its response to the radiative forcing.

The regional air temperature T hence evolves in time as follows:

$$\frac{dT}{dt} = \frac{[T_{\text{ref}} + c_I \cdot R_I + c_C \cdot R_C - c_z \cdot z] - T}{\tau_T}$$

with c_I, c_C, c_Z the climate sensitivities to insolation, atmospheric CO_2 radiative forcing and ice-sheet height, respectively,

$R_I = I - I_{\text{ref}}$

and

$R_C = 5.35 \cdot log\left(\frac{C}{C_{\text{ref}}}\right)$.

The latter was formulated by Myhre et al. (1998) and it is commonly employed in conceptual modeling.

$$\frac{dC}{dt} = \frac{[C_{\text{ref}} + k_{T,C} \cdot (T - T_{\text{ref}})] - C}{\tau_C}.$$

This reflects the capability of the sources and sinks of atmospheric CO_2 to vary with temperature.

$$\frac{dC}{dt} = \frac{[C_{\text{ref}} + k_{T,C} \cdot (T - T_{\text{ref}})] - C}{\tau_C}.$$

This reflects the capability of the sources and sinks of atmospheric CO_2 to vary with temperature.

Another improvement in the model is the ablation term, that now follows an ITM-like parameterization ature Melting method, Pellicciotti et al., 2005; Van Den Berg et al., 2008; Robinson et al., 2010)

$$\dot{a} = k_I \cdot (1 - \alpha) \cdot (I - I_{\text{ref}}) + \lambda \cdot (T - T_{\text{thr}})$$

$$\dot{s} = \dot{s}_{\text{ref}} + k_{\dot{s}} \cdot (T - T_{\text{ref}}) .$$

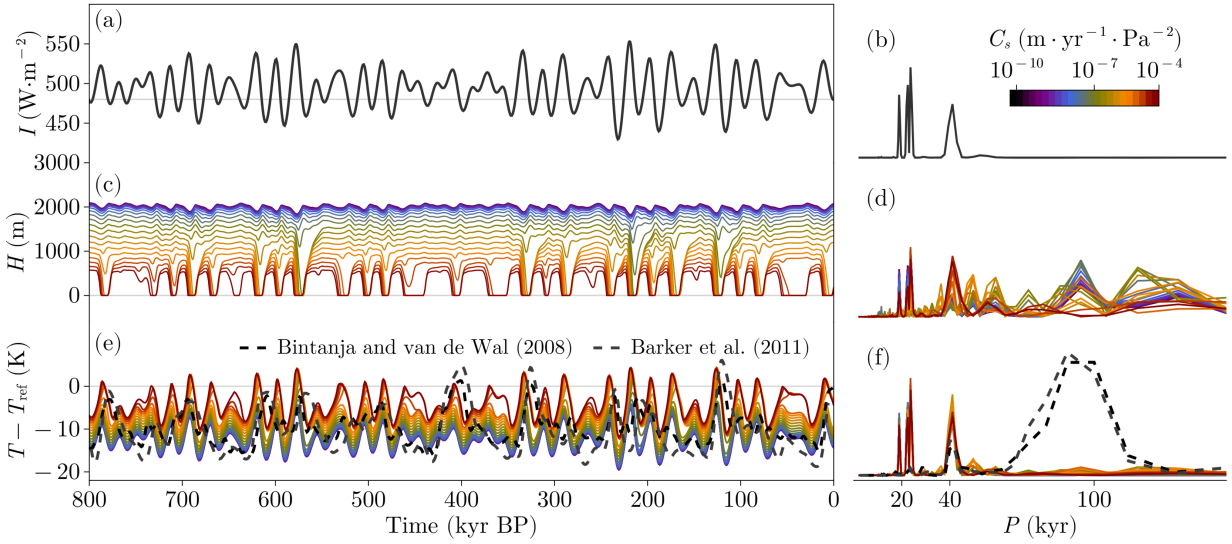


Figure 11. Results of the BASE simulation. (a, c, e) Time series obtained from the model using different sliding factors C_s. (b, d, f) Periodograms obtained from the time series in the left column. Note that when normalizing spectra, series were cut off for periods larger than 200 kyr. Note that in (e, f) dashed lines refer to two different proxies.

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> With this, we produce a more realistic 100 kyr periodicity in H than in RISOS when also using real insolation forcing. However, the timing and periodicity of T are not satisfactory yet, suggesting the nonlinear response of the model is250 still too weak to produce reliable GIV.

THERM Experiment

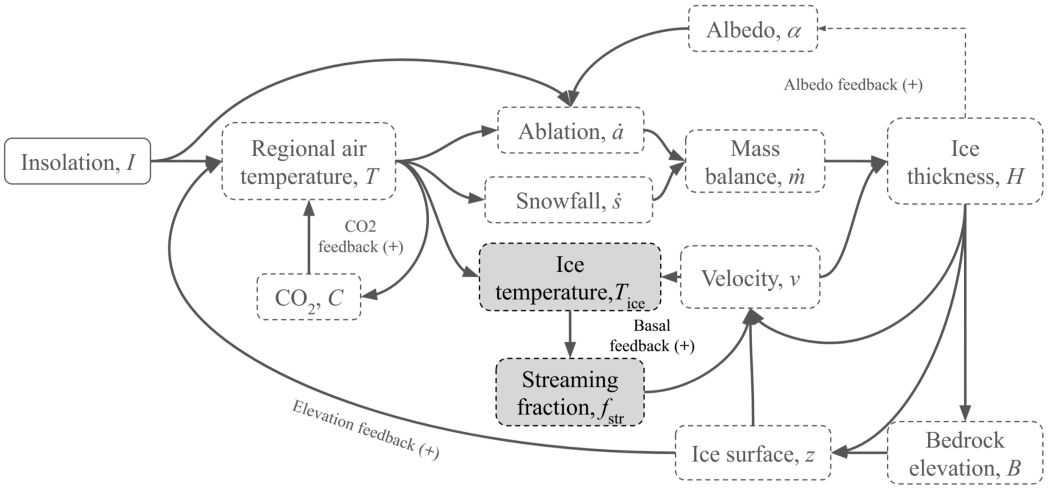


Figure 12. THERM experiment scheme. Model additions with respect to BASE are highlighted.

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> One way to enhance the nonlinear response of the system is via ice-sheet thermodynamics and its effect on the basal sliding and streaming potential of the ice sheet (i.e. thermodynamic hypothesis, Verbitsky et al., 2018; Ganopolski, 2024). The temperature of the base of an ice sheet (Tice) is influenced by both ice insulation and ice creep. If basal temperature reaches the pressure melting point, basal sliding is enhanced and ice streams accelerate and expand further.

$$\frac{dT_{\mathrm{ice}}}{dt} = \frac{1}{c_{\mathrm{ice}} \cdot \rho_{\mathrm{ice}} \cdot H_b} \cdot \frac{dh}{dt},$$

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> The bed of an ice sheet (temperate environment) is exposed to the geothermal heat flow hgeo from the lithosphere, here taken to be a constant parameter (Table 3), and, to the heat flux due to basal drag:

$$h_{\mathrm{drag}} = v_b \cdot \tau_b.$$

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These two terms act as a heat source to the ice sheet through its base. At the air-ice interface (cold environment) there is a heat loss due to ice conduction (hcond):

$$h_{\mathrm{cond}} = -k_T \cdot \nabla T_{\mathrm{ice}},$$

$$h_{\mathrm{cond}} = k_T \cdot \frac{T - T_{\mathrm{ice}}}{H}.$$

Pérez_Montero_etal_2025_pacco, p.16

Once the thermodynamics are defined, the effect on ice-sheet dynamics is translated through the fraction of ice streams fstr within the ice sheet, which evolves in time according to

$$\frac{df_{\mathrm{str}}}{dt} = \frac{f_{\mathrm{str,ref}} - f_{\mathrm{str}}}{\tau_{\mathrm{kin}}},$$

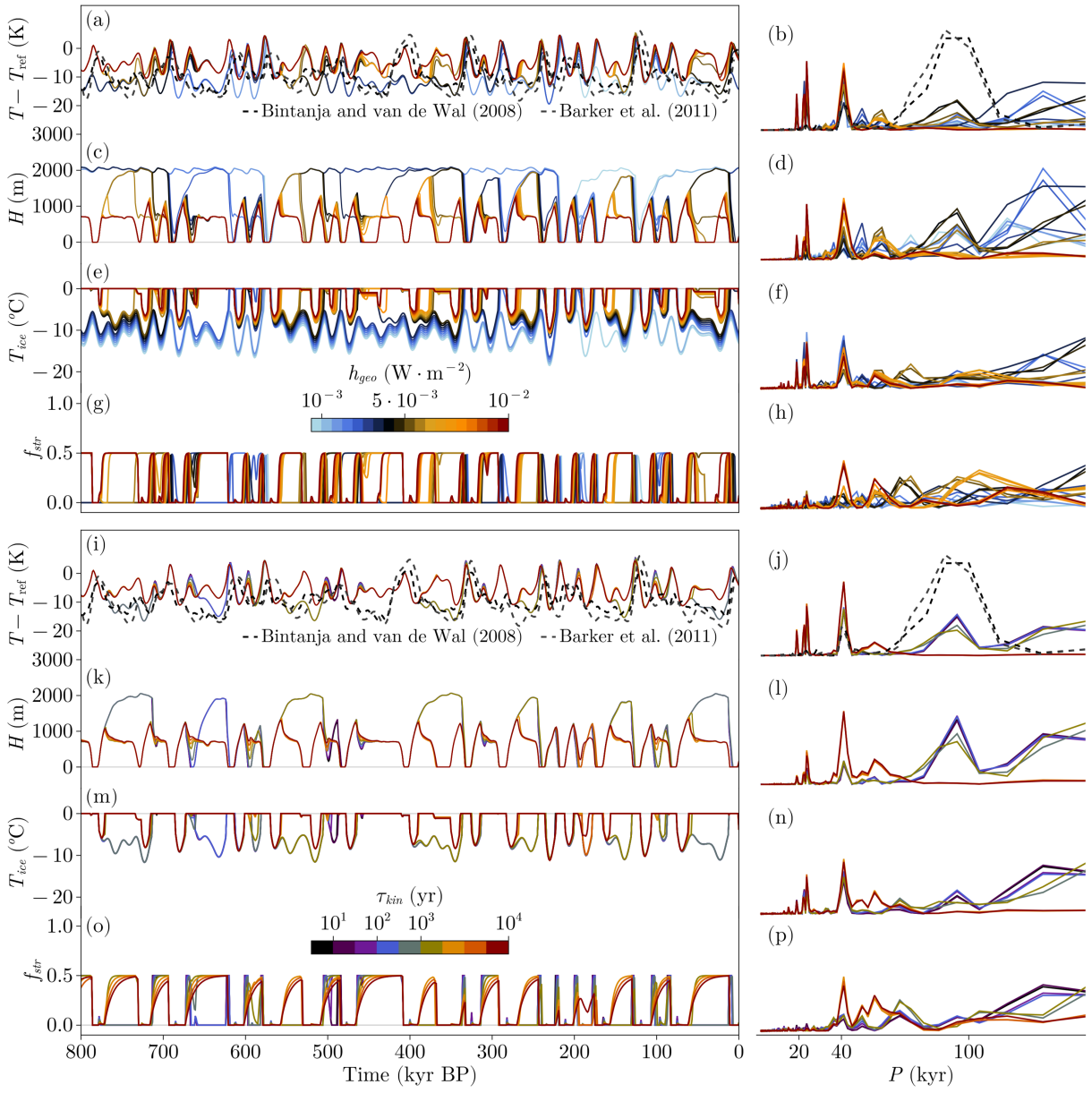


Figure 13. Results of the THERM simulation. (a, c, e, g) Time series obtained from the model using different sliding factors C_s. (i, k, m, o) Time series obtained from the model using different characteristic times, τ_{kin}. (b, d, f, h, j, l, n, p) Periodograms obtained from the time series in the left column. Note that when normalizing spectra, series were cut off for periods larger than 200 kyr.

Pérez_Montero_etal_2025_pacco, p.17

Sensitivity experiments with respect to hgeo and τkin are carried out (Fig. 13). The former shows that if hgeo is very large, the model cannot lose enough heat and the ice sheet experiences strong, frequent streaming, yielding typical periodicities below 60 kyr. Thus, H is modulated in a similar fashion to the previous purely dynamic experiments. However, for a more balanced hgeo with the rest of the components, Tice evolves more slowly and the streaming is also propagated more slowly, producing oscillations of about 80-100 kyr. In contrast, the response displays quite low sensitivity when sampling τkin within a plausible range, τkin ≤ 1000 yr, there are not many differences and the ice sheet tends to oscillate with a realistic GIV amplitude but the timing is not satisfactory. Note that in these experiments Cs was fixed to 10−5 m · yr−1 · Pa−2 since a sensitivity experiment for this parameter (not shown) showed that the purely dynamic nature of the thermodynamic hypothesis was in need of higher basal velocities. **Despite the introduction of enhanced ice dynamics via the addition of basal velocities dependent on ice thermodynamics, the THERM configuration does not capture the 100 kyr periodicity well**.

AGING Experiment

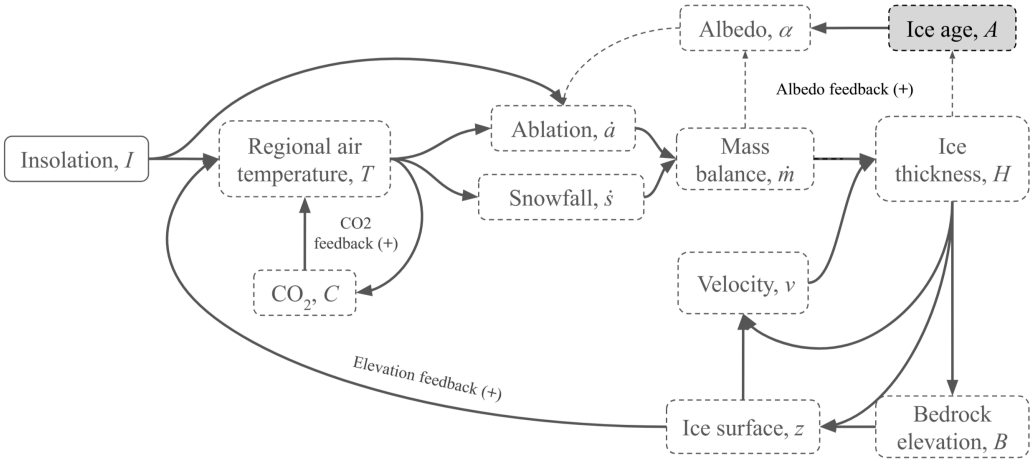


Figure 14. AGING experiment scheme. Model additions with respect to BASE are highlighted.

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Another way of amplify the nonlinear response of the Earth system to insolation forcing is through the reduction in ice albedo305 due to its natural darkening and dust deposition (Ganopolski and Calov, 2011; Willeit et al., 2019; Ganopolski, 2024). The former is related to the compaction of snow and the latter due to the aridity of glacial landscapes, which favors the deposition of dust on ice sheets. These phenomena can be easily implemented in PACCO via a relaxation equation for the albedo α:

$$\frac{d\alpha}{dt} = \frac{\alpha_{\mathrm{ref}} - \alpha}{\tau_\alpha},$$

with τα the relaxation time associated with the change in albedo for the entire ice sheet. The target value αref changes with the age of the ice sheet. For simplicity, it follows a linear relationship:

$$\alpha_{\mathrm{ref}} = \alpha_{\mathrm{ni}} - k_\alpha \cdot A,$$

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> In this manner, if the ice315 is maintained enough in time, its decreasing albedo results in a nonlinear increase of ablation. However, the assumption that aging affects the entire ice sheet is a crude one, since the accumulation areas are covered by fresh snow and, therefore, their contribution to ablation must be smaller. Thus, we add an additional constraint
>
> $$\alpha = \alpha_{\mathrm{ni}} \quad if\ \dot{m} > 0.$$

This treatment of aging is very very gross, isn't it?

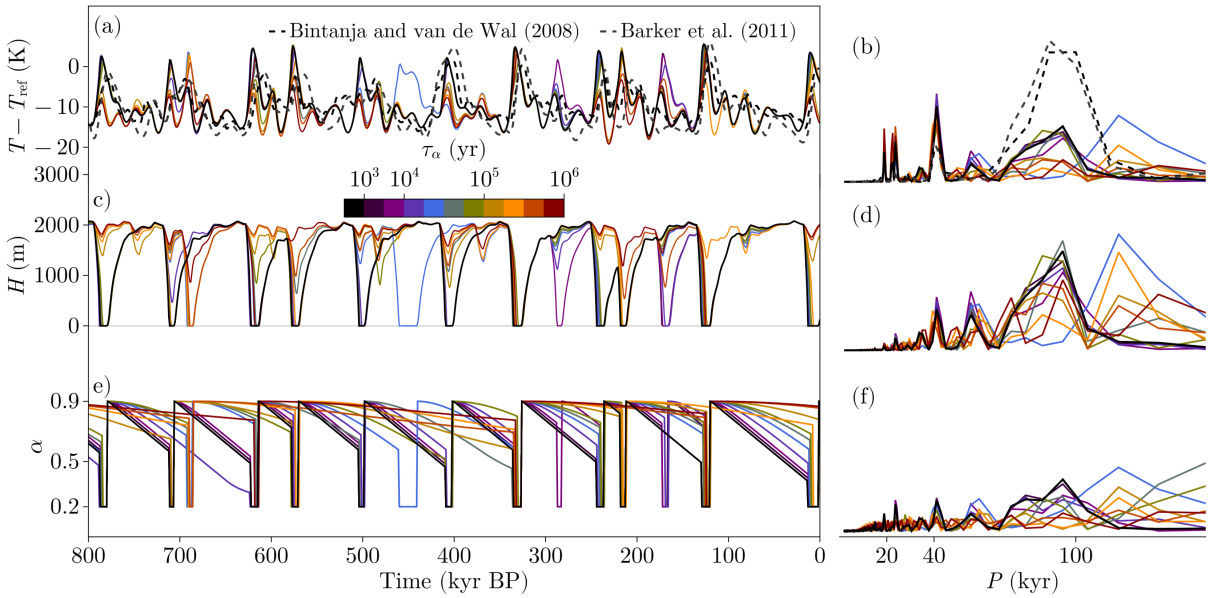


Figure 15. Results of the AGING simulation. (a, c, e) Time series obtained from the model using different aging times τ_α. (b, d, f) Periodograms obtained from the time series in the left column. Note that when normalizing spectra, series were cut off for periods larger than 200 kyr.

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> Sensitivity experiments with respect to the sliding factor Cs show that a value from 10−10 to 10−7 (range from Pollard and DeConto, 2012) produces realistic oscillations (Fig. A1). Thus, we fixed the value of Cs to 10−9 based on the idea that Late Pleistocene ice sheets were slow and not quite dynamic (Berends et al., 2021a, , Fig. 15). Our results show that the aging time τα is relevant to reproduce the correct GIV. In this experiment, 1000 yr seems to be a good value (highlighted in black in Fig. 15) since it provides high spectral power in 100 kyr band and a good agreement with proxies. Thus, we will compare the rest of the states of this model run with different proxies.

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> The evolution of Vice matches the proxy record qualitatively well despite its simple representation. In this model configuration, deglaciations start because the first term of Eq. (27) (i.e. the effective shortwave radiation Ieff , Eq. A1) increases ablation thanks to the aging in ice albedo (Fig. A2). At a certain time this contribution is high enough to outweigh snowfall, and the glacial termination starts. This process is enhanced with the ice-sheet surface elevation at the end of the glacial termination. This phenomena can be seen in Fig. 17, where we have represented the trajectories of H as a function of the insolation forcing. This diagram explicitly shows the nonlinearities in the system: ice starts to grow when insolation becomes low enough to allow its persistence over several precession and obliquity cycles, until

> the effective shortwave Ieff outweighs the accumulation. Then, a glacial termination is triggered.

Discussion

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> We have seen that the last configuration of PACCO (AGING experiment, Sect. 2.7) produces results that highly correlate with various paleoclimatic records. The mechanisms that trigger GIV in this configuration are in agreement with state-of-the-art hypotheses (Ganopolski and Calov, 2011; Abe-Ouchi et al., 2013, among others). The 100 kyr periodicity is produced due to both ice sheet dynamics and interactions with the climate. On one hand, ice-sheet velocity determines the responsiveness of the ice sheet to the insolation forcing. The change in Cs affects the level of ice sheet sliding and thus facilitates its synchronization with the forcing via ice discharge. Therefore, the periodicity of the system response varies with Cs. On the other hand, isostasy and ice aging provide two essential feedbacks that trigger lagged deglaciations. Both processes become more and more important when the ice persists in time. When the albedo is low enough, an increase in insolation can trigger a deglaciation that is reinforced with the ice surface elevation feedback. We have also seen that ice-sheet thermodynamics could play an important role in this variability since it increases the ice basal velocity. However, in our case ice aging is more effective, because it introduces a mechanism that persists over multiple glacial cycles.

Table 4. Summary and main conclusions of the experiments performed in this work.

Experiment	Conclusion
LIN	Linearly forced mass balance system behaves nonlinearly due to ice sheet dynamics
NONLIN	The system reacts at periodicities greater than the insolation forcing for moderate and weak basal ice-sheet dynamics
ISOS	Isostatic delayed rebound slows down the mass balance response of the system and facilitates terminations regardless of the intensity of basal sliding
RISOS	By forcing the system with the real insolation, resonance at greater periodicities than the forcing appears, but the very simple climate formulation does not allow for a satisfactory calibration with the real timing of the cycles
BASE	Introducing am improved ablation scheme, the effect of the ice sheet size on the regional temperature and a simplified carbon cycle allows better agreement with proxies
THERM	Including ice thermodynamics and its effects on basal sliding favors the emergence of 100 kyr periodicity but still shows a poor agreement with the terminations timing shown by the proxies
AGING	Ice darkening enhances ablation slowly and finally provides a very good simulated timing and amplitude of glacial cycles

Conclusions

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> Our model is novel in the way it is formulated because the equations related to ice dynamics are obtained through spatial lumping of ice-sheet modeling equations.

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> To our knowledge, PACCO is the first conceptual model to explicitly resolve the most important processes and to produce the Quaternary glacial-interglacial cycles with minimal physics. It opens a new way of conceptual modeling that allows different hypotheses to coexist and be isolated from one another.

DOUBTS

- What offers this model to the modelling community? Why to use this model and no other for my TFM validation?
- How to pass from a dimensional to an adimensional model, like PACCO. Please, show me the real equations and then the decisions taken to derive the adimensional model.
- Why always C_s variable? Is it because we don't know it's real value?
- Why did you give me BASE simulations if they do not capture100 kyr variability? And why not, if they have delayed isostatic rebound? Abe-Ouchi, 2014, says that with it, 100 kyr cycles must be reproduced.
- Why so large V_ice equivalent to sea level in BASE.nc? 150 m??? And why we don't see postive values?